

Pattern Recognition MCQs and Fill in the Blanks - Answered

Correct answers are highlighted in yellow. Short notes are added only where the question is ambiguous or commonly confused.

A. Multiple Choice Questions (MCQs)

1. What is the main difference between parametric and nonparametric learning methods?

(a) Parametric methods assume a specific form for the underlying distribution, while nonparametric methods do not

- (b) Parametric methods are faster for large datasets
- (c) Nonparametric methods require fewer training samples
- (d) Parametric methods are always more accurate

Answer: (a) Parametric methods assume a specific form for the underlying distribution, while nonparametric methods do not

2. Which of the following is a nonlinear discriminant function?

(a) Quadratic Discriminant Analysis (QDA)

- (b) Fisher's Linear Discriminant
- (c) Logistic Regression
- (d) Single perceptron model

Answer: (a) Quadratic Discriminant Analysis (QDA)

Note: QDA gives quadratic decision boundaries, so it is nonlinear.

3. When you find noise in data, which option would you consider in k-NN classification?

(a) Value of k can be increased

- (b) Value of k can be decreased
- (c) Noise cannot be dependent on value of k
- (d) None of (a), (b) and (c)

Answer: (a) Value of k can be increased

Note: Increasing k smooths the decision boundary and reduces sensitivity to noisy points.

4. Removing duplicate records is a process called

- (a) Recovery
- (b) Data cleaning**
- (c) Data dredging
- (d) Data pruning

Answer: (b) Data cleaning

5. Which of the following is a nonparametric learning algorithm?

(a) k-Nearest Neighbors

- (b) Linear Discriminant Analysis
- (c) Logistic Regression
- (d) Single perceptron model

Answer: (a) k-Nearest Neighbors

6. Which of the following measure is not a metric?

- (a) Euclidean Distance
- (b) Cosine Similarity
- (c) KL divergence**
- (d) None of these

Answer: (c) KL divergence

Note: Strictly, cosine similarity is also not a distance metric because it is a similarity measure. If the exam expects one option, KL divergence is the standard non-metric answer because it is not symmetric and does not satisfy triangle inequality.

7. Which of the following is a metric distance measure?

- (a) Euclidean Distance

- (b) Manhattan Distance
- (c) Minkowski Distance

(d) All of the above

Answer: (d) All of the above

8. In feature selection, what does a low p-value in One-Way ANOVA indicate?

(a) The feature is likely to be significant for classification

- (b) The feature is irrelevant for classification
- (c) The feature has no variance
- (d) The feature is highly correlated with other features

Answer: (a) The feature is likely to be significant for classification

9. Choose the general limitations of the backpropagation rule among the following.

- (a) Slow convergence
- (b) Scaling
- (c) Local minima problem

(d) All of the above

Answer: (d) All of the above

10. Which guideline is applicable to initialization of the weight vector in a fully connected neural network?

- (a) Should not set it to zero since otherwise it will cause overfitting
- (b) Should not set it to zero since otherwise stochastic gradient descent will explore a very small space**
- (c) Should set it to zero since otherwise it causes a bias
- (d) Should set it to zero in order to preserve symmetry across all neurons

Answer: (b) Should not set it to zero since otherwise stochastic gradient descent will explore a very small space

Note: Zero initialization causes symmetry; neurons learn the same features, so training becomes ineffective.

11. The numerical output of a sigmoid node in a neural network

- (a) Is unbounded, encompassing all real numbers
- (b) Is unbounded, encompassing all integers
- (c) Is bounded between 0 and 1**
- (d) Is bounded between -1 and 1

Answer: (c) Is bounded between 0 and 1

12. Statement 1: The error surface followed by the gradient descent backpropagation algorithm changes if we change the training data. Statement 2: Stochastic gradient descent is always a better idea than batch gradient descent.

(a) Only statement 1 is true

- (b) Only statement 2 is true
- (c) Both are false
- (d) Both are true

Answer: (a) Only statement 1 is true

Note: Statement 2 is false because SGD is not always better; choice depends on data size, noise, stability and convergence needs.

13. Gain Ratio has an advantage over Gain of an attribute when the

- (a) Attribute is categorical with 3 possible values
- (b) Attribute has multiple values, and each is unique**
- (c) Attribute is numerical
- (d) Numerical with large number of duplicate values

Answer: (b) Attribute has multiple values, and each is unique

Note: Gain Ratio reduces the bias of Information Gain toward attributes with many distinct values.

14. Single Perceptron classifier is not able to implement

- (a) OR gate
- (b) AND gate
- (c) XOR gate**

(d) NOT gate

Answer: (c) XOR gate

Note: XOR is not linearly separable.

15. The Gini impurity function is maximum in a binary classified dataset when prob(Class 1) is

(a) 1.00

(b) 0.25

(c) 0.50

(d) 0

Answer: (c) 0.50

16. Suppose that $X_1, \dots, X_m$ are categorical input attributes and Y is a categorical output attribute. Suppose we plan to learn a decision tree without pruning, using the standard algorithm. The maximum depth of the decision tree must be

(a) Less than $m+1$

(b) Greater than $m+1$

(c) Both (a) and (b) can be true

(d) Neither (a) nor (b) is true

Answer: (a) Less than $m+1$

Note: A path can test at most m attributes, so the edge-depth is at most m , i.e., less than $m+1$.

17. Which of the following is a limitation of Expectation Maximization Clustering?

(a) Flexible Cluster Shapes

(b) Soft Assignment

(c) Handles Overlapping Data

(d) Computational Complexity

Answer: (d) Computational Complexity

18. In which step does the Expectation Maximization Clustering algorithm calculate the probability that each data point belongs to each cluster based on the current parameter estimates?

(a) Maximization

(b) Minimization

(c) Expectation

(d) None of the above

Answer: (c) Expectation

19. Expectation Maximization Clustering uses a _____ to cluster the data.

(a) Probabilistic approach

(b) Non-probabilistic approach

(c) Arithmetic approach

(d) Algebraic approach

Answer: (a) Probabilistic approach

20. Hierarchical agglomerative based clustering is a

(a) Bottom up approach

(b) Top down approach

(c) Both (a) and (b)

(d) None of the above

Answer: (a) Bottom up approach

21. In DBSCAN algorithm, points having at least minimum required neighbours are

(a) Core points

(b) Border points

(c) Both (a) and (b)

(d) None of the above

Answer: (a) Core points

22. DBSCAN cannot be used with high accuracy for datasets that are

- (a) Convex
- (b) Uniform density
- (c) Non-uniform density**
- (d) None of (a), (b) and (c)

Answer: (c) Non-uniform density

Note: DBSCAN struggles when clusters have very different densities because one epsilon value may not suit all clusters.

23. A density based clustering algorithm is

- (a) DBSCAN**
- (b) FCM
- (c) k-means
- (d) Agglomerative

Answer: (a) DBSCAN

24. In Fuzzy C Means, C represents

- (a) Number of data points
- (b) Number of clusters**
- (c) Number of border points
- (d) Number of neighbours

Answer: (b) Number of clusters

25. When performing regression or classification, which is the correct way to pre-process the data when PCA is used?

- (a) Normalize the data -> PCA -> training**
- (b) PCA -> normalize PCA output -> training
- (c) Normalize the data -> PCA -> normalize PCA output -> training
- (d) None of the above

Answer: (a) Normalize the data -> PCA -> training

Note: Standard PCA preprocessing is to scale/normalize features first, then apply PCA, then train the model on principal components.

26. If the variance-covariance matrix is a diagonal matrix, then the features are _____.

- (a) Independent**
- (b) Dependent
- (c) Zero mean value
- (d) Uniformly distributed

Answer: (a) Independent

Note: More precise statistical wording: the features are uncorrelated. For multivariate Gaussian data, uncorrelated implies independent; hence option (a) is expected.

27. Consider the statements: Statement 1: Principal Component Analysis is an unsupervised method. Statement 2: It searches for the directions that data have the largest variance.

- (a) Only Statement 1 is correct
- (b) Only Statement 2 is correct
- (c) Both statements are incorrect
- (d) Both statements are correct**

Answer: (d) Both statements are correct

28. Consider the statements: Statement 1: Independent Component Analysis is an unsupervised method. Statement 2: Independent Component Analysis works very well with Gaussian data distribution.

- (a) Only Statement 1 is correct**
- (b) Only Statement 2 is correct
- (c) Both statements are incorrect
- (d) Both statements are correct

Answer: (a) Only Statement 1 is correct

Note: ICA requires non-Gaussian independent sources; it fails or becomes non-identifiable for Gaussian sources.

29. Consider the statements: Statement 1: Linear Discriminant Analysis can be used in data classification. Statement 2: Linear Discriminant Analysis can be used in dimension reduction.

- (a) Only Statement 1 is correct
- (b) Only Statement 2 is correct
- (c) Both statements are incorrect
- (d) Both statements are correct**

Answer: (d) Both statements are correct

30. Which technique would perform better for reducing the dimensions of a dataset?

- (a) Removing columns that have too many missing values**
- (b) Removing columns that have high variance in data
- (c) Removing columns with dissimilar data trends
- (d) None of these

Answer: (a) Removing columns that have too many missing values

Note: Removing columns with too many missing values is a valid preprocessing-based dimensionality reduction step. Do not remove high-variance columns just because they have high variance.

31. What will happen when eigenvalues are roughly equal?

- (a) PCA will perform outstandingly
- (b) PCA will perform badly**
- (c) Can't say
- (d) None of above

Answer: (b) PCA will perform badly

Note: If eigenvalues are roughly equal, no principal component captures much more variance than others, so PCA gives weak reduction.

32. Which of the following is true about LDA?

- (a) LDA aims to maximize the distance between classes and minimize the within-class distance**
- (b) LDA aims to minimize both distances between classes and the distance within the class
- (c) LDA aims to minimize the distance between classes and maximize the distance within the class
- (d) LDA aims to maximize both distances between classes and the distance within the class

Answer: (a) LDA aims to maximize the distance between classes and minimize the within-class distance

B. Fill in the Blanks

1. In a decision tree node for binary class, the probability of one class is 0.75. The entropy of the node is Answer: **0.811 bits (approximately)**
2. _____ probability distribution is often used in Linear discriminant function. Answer: **Gaussian / Normal**
3. _____ model use Bayes rule for classification tasks. Answer: **Bayesian classification / Naive Bayes**
4. The null hypothesis in the Chi-Square test for feature selection states that the feature and the target variable are _____ with respect to each other. Answer: **independent**
5. Fisher's linear discriminant analysis is used in _____. Answer: **classification and dimensionality reduction**
6. The objective of a good clustering technique is to minimize _____ distance between clusters. Answer: **intra-cluster / within-cluster**
7. The objective of a good clustering technique is to maximize _____ distance between clusters. Answer: **inter-cluster / between-cluster**
8. In a decision tree node the number of data items of class C1, class C2 and class C3 are 14, 25 and 11 respectively then the Gini impurity of the node will be Answer: **0.6232**
9. In a decision tree node the number of data items of class C1, class C2 and class C3 are 14, 25 and 11 respectively then the entropy of the node will be Answer: **1.4948 bits (approximately)**
10. In a decision tree node the number of data items of class C1, class C2 and class C3 are 14, 25 and 11 respectively then the misclassification error based impurity of the node will be Answer: **0.50**
11. When two classes can be separated by a separate line, they are known as Answer: **linearly separable**
12. Impurity of decision tree node with respect to an attribute is high means this attribute has _____ effect in classification. Answer: **less / weak**
13. In L1 regularization (Lasso), the coefficients of irrelevant features will shrink to Answer: **zero**
14. If probability of heads is 0.9 then what is the probability of generating the sequence 'hhhhth' using six independent tosses, then $P(\text{hhhhth})$ is Answer: **0.059049**
15. _____ is soft clustering algorithm. Answer: **Fuzzy C-Means (FCM)**
16. _____ function is used in the output layer of a MLP classifier used for classifying multi class dataset. Answer: **Softmax**

C. Quick Calculation Notes for Numeric Fill in the Blanks

Entropy for $p = 0.75$: $H = -0.75 \log_2(0.75) - 0.25 \log_2(0.25) = 0.811$ bits
Gini impurity for counts 14, 25, 11: Total = 50; $Gini = 1 - [(14/50)^2 + (25/50)^2 + (11/50)^2] = 0.6232$
Entropy for counts 14, 25, 11: $H = -\sum p_i \log_2(p_i)$, where $p = 0.28, 0.50, 0.22$; $H = 1.4948$ bits
Misclassification error for counts 14, 25, 11: Error = $1 - \max(14/50, 25/50, 11/50) = 1 - 0.5 = 0.50$
Coin toss sequence hhhth: $P = 0.9^5 \times 0.1 = 0.059049$